

Nguyen Hua Gia Phuc

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github.com/hgpnguyen • personal-portfolio-orcin-rho-95.vercel.app

Software Engineer with a Master's in Information Systems and 2+ years of experience building high-performance backend systems. Expert in Python and Java, with a proven track record of deliver high-quality projects. Proficient in **Django** for web applications, with strong database management skills

WORK EXPERIENCE

Research Assistant

Nov 2025 - May 2026

Loma Linda University

Loma Linda, USA

- **Architected** modular data pipelines in **Python** to preprocess and analyze high-dimensional radiomic datasets for survival rate prediction.
- **Developed and validated** predictive machine learning models using **Scikit-learn**, implementing robust feature selection and cross-validation frameworks
- Implemented robust feature selection algorithms and cross-validation frameworks in **Python**, translating complex clinical hypotheses into scalable, high-performance code.

Software Engineer

Apr 2025 - Oct 2025

KTest JSC

Ho Chi Minh City, Viet Nam

- **Engineered** a scalable backend for an AI-driven legal-tech platform using **Python** and **Django**, handling complex regulatory queries.
- **Designed** automated document-parsing logic and state-management systems to facilitate high-accuracy data entry for legal filings.
- **Deployed** containerized services using **Docker**, streamlining the development-to-production pipeline.

Research Engineer

Feb 2023 - Aug 2024

SCIS Singapore Management University

Singapore

- Researched and developed method to generate **programming exercises** using **Python** and **OpenAI**, resulting in a comprehensive database of over **100 exercises** tailored for beginner students.
- Leveraged **Flask** and **TypeScript** to create a **Visual Studio Code extension** that enhanced student engagement and improved exercise generation efficiency by **30%**.
- This work culminated in a **published research paper** and a **practical tool for learning**.

SKILLS

Programming language : Python(advance), Java, C++, TypeScript, JavaScript

AI & Data: PyTorch, Scikit-learn, OpenAI API, NLP Pipelines

Database: SQL Server, MySQL, PostgreSQL

Testing: Pytest, JUnit

Backend & Cloud: Django, Flask, Node.js, Spring Boot, Docker, Git, Linux, AWS, Azure

PROJECTS

LittleLemon

Jan 2025 - Feb 2025

A **restaurant API** developed with role-based access for **Customers, Managers, and Delivery Crew**. Customers can view the menu, manage cart, and place orders. Managers can manage menu items, orders, and user roles. Delivery crew can update order status. Results are paginated, with filtering and search capabilities.

OpenWeather API

Nov 2024 - Dec 2024

A **CLI Python** application that retrieves weather data from the *OpenWeather API* for a specified region, optimized through parallel API calls to efficiently collect data from multiple locations. The user inputs a bounding box to define the region and retrieve all weather data in that region. A detailed **README** is included, outlining setup, usage, and the optimization approach.

Task Manager

Nov 2024 - Dec 2024

A *Task Management API* developed enables users to create, update, delete, and view tasks, each with a title, description, due date, and status, stored in **PostgreSQL**. Users can retrieve all tasks or filter them by status, with results returned in paginated format. A detailed **README** provides instructions for setting up **PostgreSQL**, configuring the server, usage, and design decisions.

EDUCATION

Master of Philosophy in Information Systems

Jan 2020 - Jan 2023

Singapore Management University

GPA: 3.03/4

Attended a Master's program on scholarship and worked as a teaching assistant for a professor.

Bachelor of Engineer in Computer Science

Jan 2014 - Jan 2019

Ho Chi Minh University of Technology

GPA: 8.23/10

Volunteered for **30 days** to build road in countryside during the summer of 2016.

LANGUAGES

English, advanced (IELTS 7.0) (Professional working proficiency) • **Japanese, intermediate (N3)** (Limited working proficiency)

CERTIFICATIONS

IBM Full Stack Software Developer  by IBM

Feb 2026

PUBLICATIONS

A CNN Model with Data Imbalance Handling for Course-Level Student Prediction Based on Forum Texts. Nguyen Hua Gia Phuc and Vo Thi Ngoc Chau.
on ICCCI 2018

ExGen: Ready-To-Use Exercise Generation in Introductory Programming Courses Nguyen Binh Duong TA, Nguyen Hua Gia Phuc & Swapna GOTTIPATI.
on ICCE 2023

REFERENCES

Chi T. Viet (Associate Professor in Loma Linda) • cviet@llu.edu

TA Nguyen Binh Duong (Associate Professor of Computer Science) • donta@smu.edu.sg